Chapter 7, Problem 9RQ

Problem

If *vs* changes from 2 V to 4 V at $t = 0$, we may express *vs* as:

(a) $\delta(t)$ V

(b) $2u(t)$ V

(c) $2u(-t) + 4u(t)$ V

(d) $2 + 2u(t)$ V

(e) $4u(t) - 2$ V

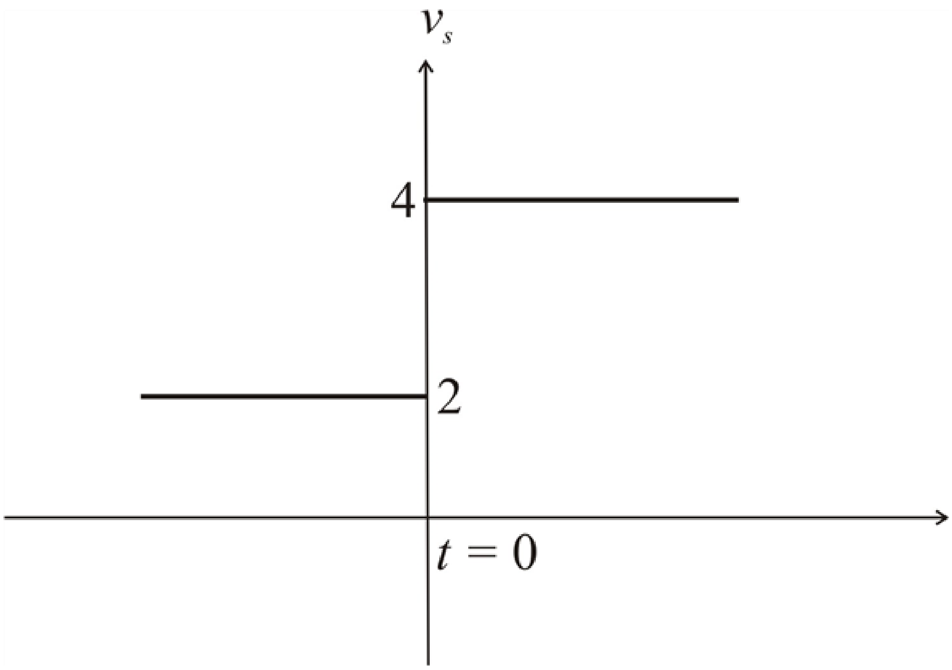
Step-by-step solution

Step 1 of 6

Given data,

The voltage, v_s changes from $2\text{ V to }4\text{ V}$ at $t = 0$

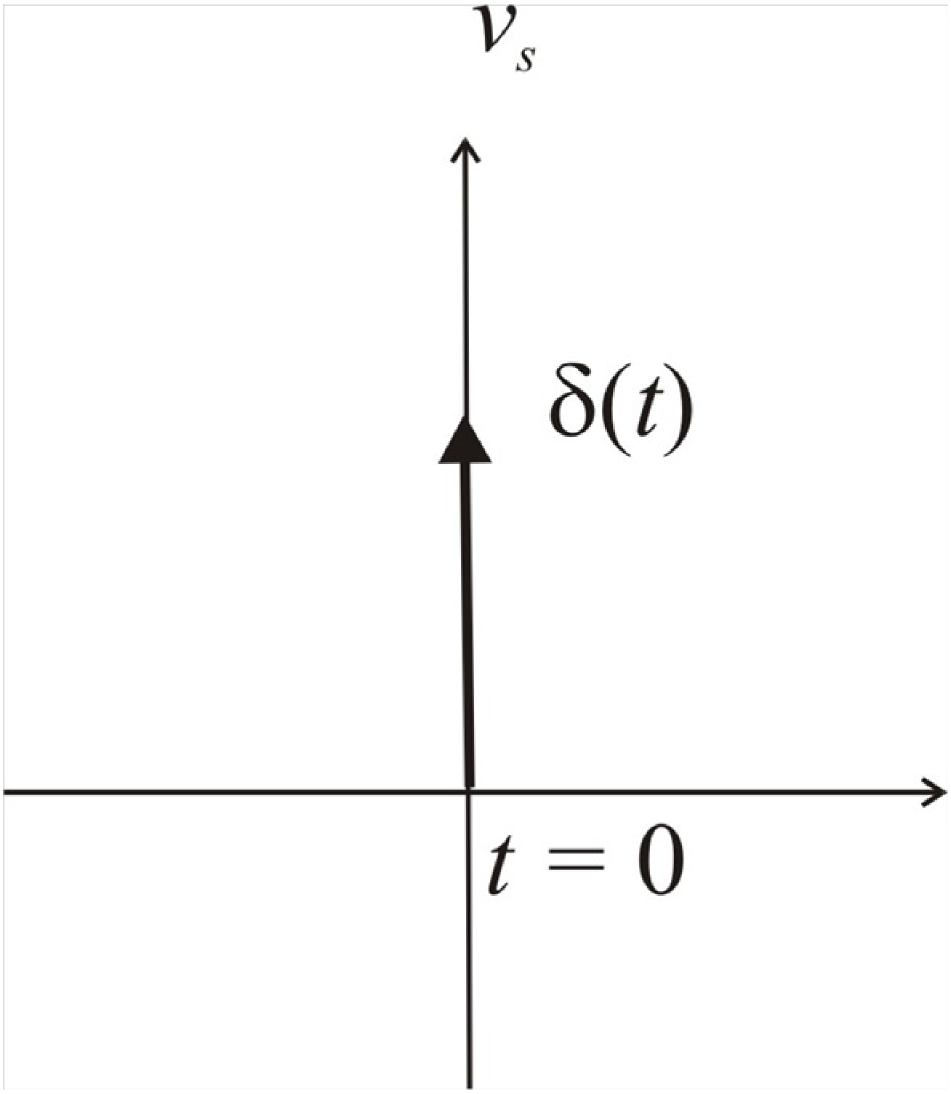
Form the given data, the waveform can be drawn as



Step 2 of 6

(a)

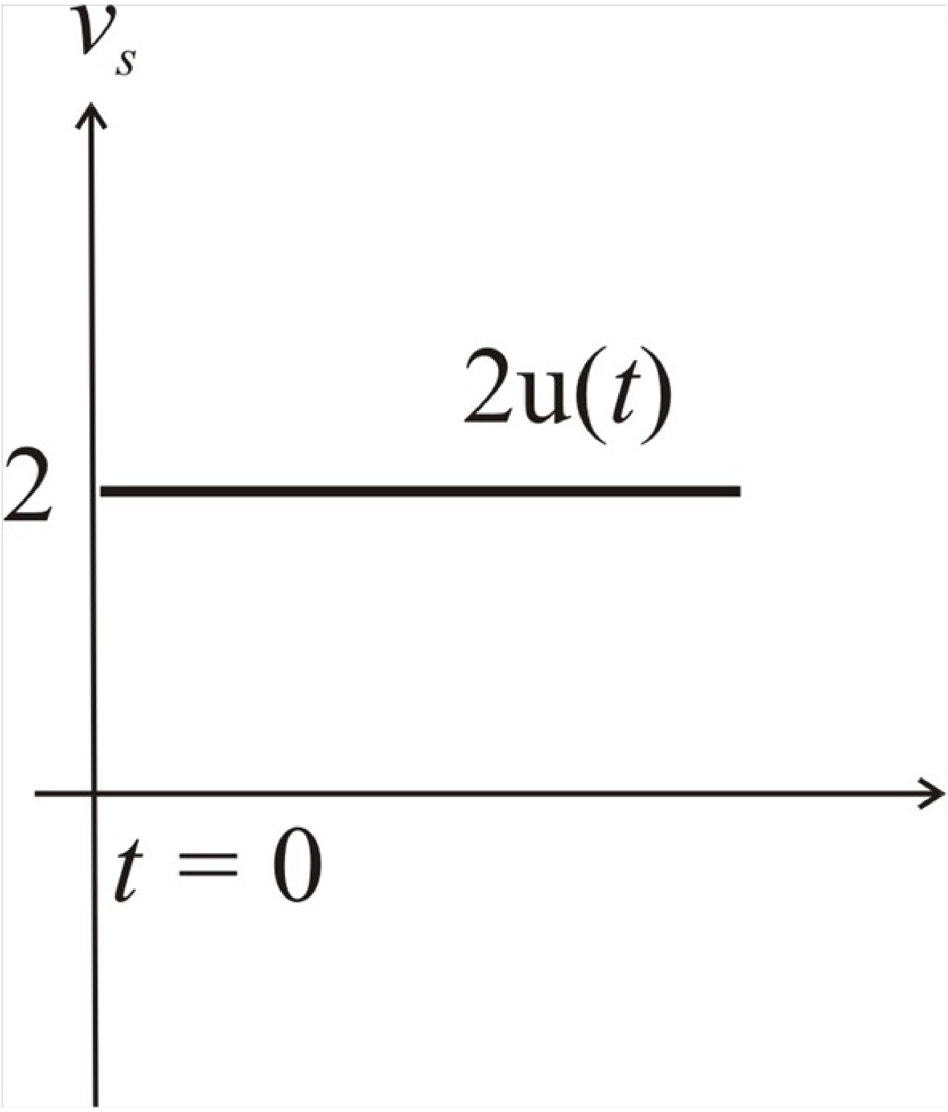
$\delta(t)$ V waveform becomes



Step 3 of 6

(b)

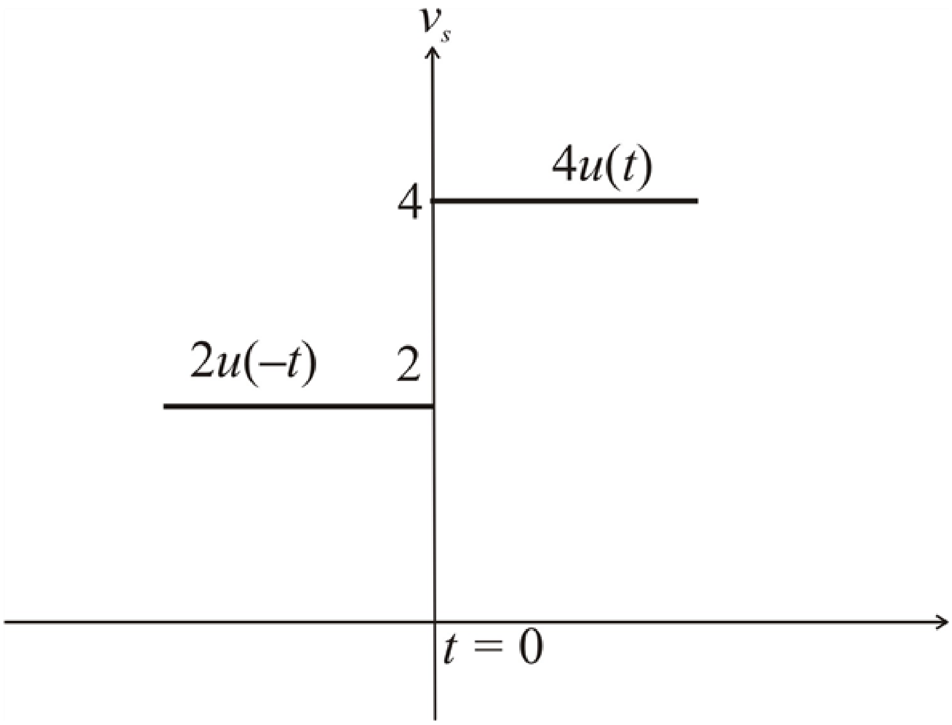
$2u(t)\,\mathrm{V}$ waveform becomes



Step 4 of 6

(c)

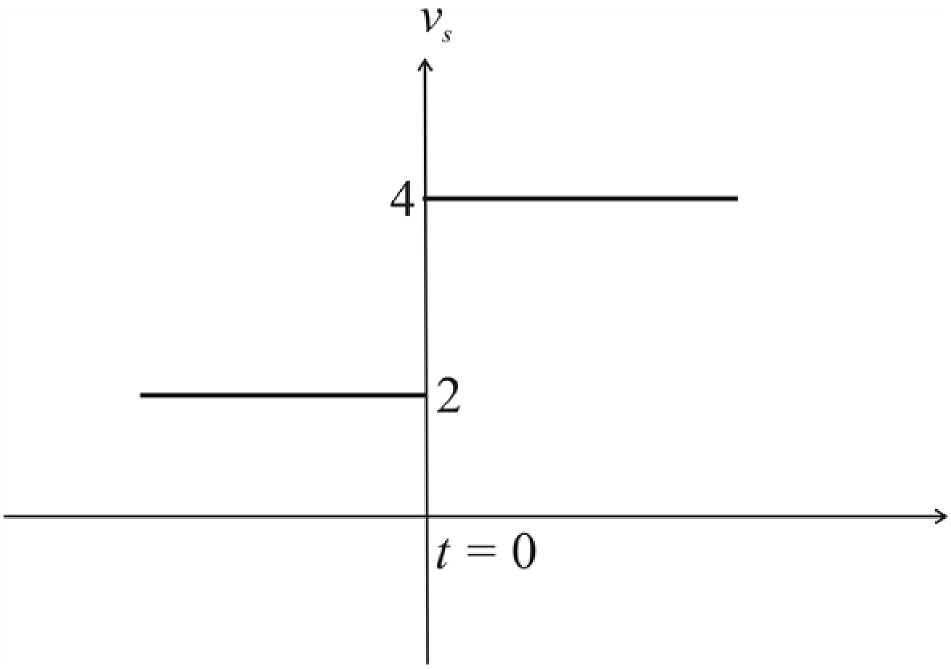
$2u(-t)+4u(t)\,\mathrm{V}$ waveform becomes



Step 5 of 6

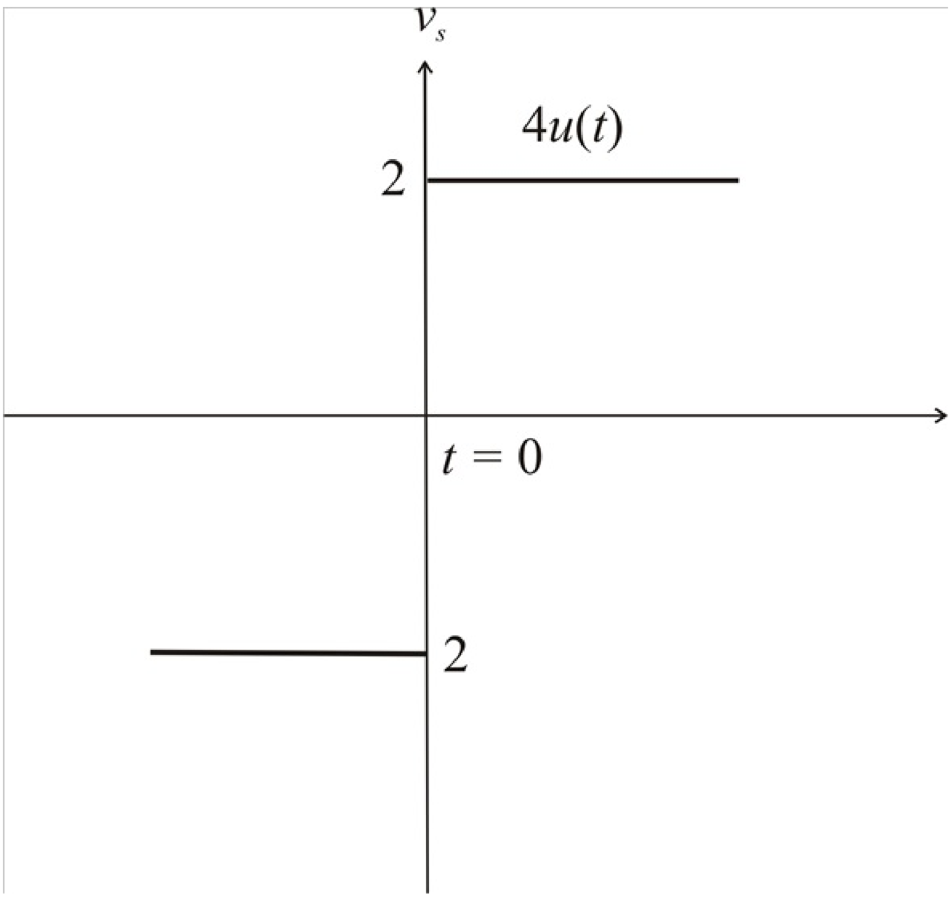
(d)

$2+2u(t)\,\mathrm{V}$ waveform becomes



(e)

$4u(t)-2\,\mathrm{V}$ waveform becomes



Step 6 of 6

The waveforms (**c**) and (**d**) matches the given waveform